ANALYTICAL REPORT

Tier 3 Demand Forecasting: Facebook Prophet with Nigerian Holiday Regressors

Procurement Demand Forecasting Initiative

EHA Clinics · Supply Chain Analytics

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This report documents the implementation and evaluation of Tier 3 (Facebook Prophet) forecasting models with Nigerian holiday and procurement-cycle regressors across all 15 eligible category-facility pairs. It presents the Q3 MAPE assessment, the improvement achieved over the Tier 1 & 2 baseline, and identifies which pairs have met the KPI target of MAPE $\leq 20\%$, along with those requiring escalation to Tier 4 (machine learning methods).

Executive Summary

This report presents the results of the Tier 3 forecasting phase of the EHA Clinics Procurement Demand Forecasting Initiative. Facebook Prophet — a decomposable Bayesian time-series model — was applied to all **15 eligible category-facility pairs** with a purpose-built Nigerian holiday and procurement-cycle regressor calendar. Models were trained on the January 2021–December 2023 window, hyperparameters were optimised via a 9-combination walk-forward cross-validation grid, and final performance was evaluated against January–June 2024 actuals.

Seven key conclusions emerge from this evaluation:

1. **Prophet materially outperforms Tier 1 and Tier 2 models on the majority of pairs.** Its explicit Fourier-series seasonality decomposition and holiday-effect regression address the primary structural weaknesses identified in classical models (SARIMA instability, ETS rigidity on multiplicative seasonal patterns).

2. **The Harmattan season regressor (Nov–Feb) is the single most impactful domain-specific feature.** OTC Drugs and Prescription Medications at anchor facilities show a clear demand elevation during the Harmattan dry season that Tier 2 models partially captured but Prophet explicitly encodes.

3. **Nigerian Islamic holiday regressors (Eid al-Fitr, Eid al-Adha) capture procurement pause-and-restock patterns.** These two events create operational disruptions — reduced ordering in the preceding weeks and elevated restocking in the following month — that Prophet's holiday window mechanism can account for.

4. **Year-end budget-flush regressors (Nov–Dec) improve accuracy on categories with programme or budget-driven procurement spikes.** Laboratory Consumables and Medical Consumables at anchor facilities both show December procurement elevation consistent with fiscal year-end spending.

5. **Hyperparameter tuning reveals that sparse, intermittent series benefit from low changepoint flexibility** (changepoint prior scale ≤ 0.05), while categories with clear trend evolution (Rx Medications at Asba) benefit from higher flexibility (scale ≥ 0.1).

6. **Laboratory Consumables remains a special case.** The all-zero validation window at Asba is a known data gap that prevents MAPE evaluation. Kano Lamido's extreme bulk-event pattern yields high error regardless of model, pending supply-chain-level root cause resolution.

7. **Pairs that still do not meet the $\leq 20\%$ MAPE target after Tier 3 are escalated to Tier 4 (XGBoost / Random Forest)** with a recommendation

to incorporate consultation-volume and external demand-driver features before model fitting.

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1. Introduction

1.1 Purpose and Context

This report is the third in a series documenting the EHA Clinics Procurement Demand Forecasting Initiative. It follows:

- **Forecasting Scope and Modelling Approach** (Concept Note, April 2026) — which defined the tiered modelling strategy, six pilot facilities, nine commodity categories, and the KPI target of MAPE $\leq 20\%$.
- **Exploratory Data Analysis Report** (April 2026) — which characterised demand patterns, identified the Kano Independence Road bulk stock-load anomaly (September 2021), and classified pairs by data readiness (Tier A/B/C).
- **Tier 1 & 2 Demand Forecasting Report** (May 2026) — which established the Q2 MAPE baseline across five classical models, found that no pair met the $\leq 20\%$ target, and escalated all 15 eligible pairs to Tier 3 (Prophet).

The present document reports the **Tier 3 evaluation**: Facebook Prophet with a domain-specific Nigerian holiday and procurement-cycle regressor calendar, systematic hyperparameter tuning via walk-forward cross-validation, and final hold-out validation against the January–June 2024 window.

1.2 Why Prophet at Tier 3

The Tier 1 and Tier 2 evaluation identified four structural failure modes in classical models that Prophet is specifically designed to address:

Failure mode (Tier 1/2)	Prophet's response
SARIMA instability on sparse series (MAPEs >1,000% on 4 pairs)	Piecewise linear trend with automatic changepoint detection requires no differencing; never produces explosive forecasts on series with zero-valued gaps.
ETS / Theta miss Q4 Harmattan OTC peak	Yearly Fourier seasonality captures amplitude and phase; a binary Harmattan regressor (Nov–Feb) adds an explicit demand-elevation term that ETS's additive seasonal component underestimates.
Vaccines miss immunisation programme procurement cycle timing	Nigerian public holidays and programme event dates are passed as Prophet holiday regressors, allowing the model to attribute procurement spikes to their causal event rather than treating them as unexplained noise.
100% MAPE floor on intermittent series (model predicts non-zero, actual is zero)	Prophet's additive regression framework with zero-clipping of forecasts handles sparsity more gracefully than ARMA-family models; the holiday window mechanism can suppress forecast elevations in months where low procurement is systematically expected.

1.3 KPI and Evaluation Protocol

The KPI from management remains unchanged:

> *"Achieve MAPE $\leq 20\%$ for critical health commodities across pilot facilities by Q3 2026."*

MAPE is computed only over months with non-zero actual procurement (to avoid division by zero on sparse series):

$$\text{MAPE} = \frac{1}{n_{\text{nz}}} \sum_{t\,:\,A_t>0} \left| \frac{A_t - F_t}{A_t} \right| \times 100$$

where n_{nz} is the count of non-zero validation months. All forecasts are clipped to zero before MAPE computation to prevent negative forecasts from inflating error.

The evaluation protocol is identical to Tier 2:

- **Walk-forward CV** (Fold 1: Jan 2021–Dec 2022 train / Jan–Jun 2023 validate; Fold 2: Jan 2021–Jun 2023 train / Jul–Dec 2023 validate) — used for hyperparameter selection.
- **Final hold-out** — full training window (Jan 2021–Dec 2023) fitted with best hyperparameters per pair; validated against Jan–Jun 2024 actuals.

2. Data Preparation

2.1 Filtering Pipeline

The Tier 3 analysis applies the identical filtering pipeline as Tier 2, ensuring full comparability of MAPE results:

1. Retain only `state = done` records.
2. Require `product_qty > 0` and `price_subtotal` ≥ 0.
3. Restrict to nine in-scope commodity categories and six pilot facilities.
4. Exclude Kano Independence Road 2021 bulk event (365 records, Sep 2021 stock-load exercise: 258,470 units).
5. Aggregate to monthly totals per category-facility pair and zero-fill all months within Jan 2021–Jun 2024, yielding a complete 42-month matrix.

Final dataset: **8,458 records** across **6 facilities**, **9 categories**, covering **45 pairs** $\times$ **42 months**.

2.2 Pair Classification

The Tier A/B/C readiness classification is carried forward from Tier 2:

Tier	Criterion	Pairs	Treatment
A	≥24 non-zero months	13	Full Prophet pipeline
B	12–23 non-zero months	2	Full Prophet pipeline (same as Tier A; Prophet handles short series robustly due to Bayesian priors)
C	<12 non-zero months	28	Excluded; fixed-interval ordering continues to apply

15 pairs proceed to Tier 3 modelling (13 Tier A + 2 Tier B).

3. Nigerian Holiday & Procurement Event Calendar

3.1 Rationale for Event Regressors

A fundamental insight from the EDA was that demand at EHA Clinics is **event-driven** as much as it is cyclically seasonal. Three distinct event mechanisms were identified:

1. **Nigerian public holidays** — most critically, Eid al-Fitr and Eid al-Adha, which create 3–5 day operational pauses at facilities serving Kano (predominantly Muslim population). Procurement orders placed immediately before these holidays spike as facilities stock up; order activity drops to near-zero during the holiday window itself.

2. **The Harmattan dry season** (November through February) is Nigeria's arid, dusty, and cold-night season. It systematically elevates demand for respiratory medications (salbutamol, antihistamines), dermatological products, and OTC analgesics. This is a recurring, predictable demand elevation that is *not* captured by standard yearly Fourier seasonality, which distributes amplitude smoothly across the calendar.

3. **Year-end fiscal-cycle procurement** (November–December) reflects EHA Clinics' budget calendar. Uncommitted supply-chain budget is often consumed in Q4, producing a consistent procurement surge in November and December that is unrelated to patient demand but is a real and predictable procurement pattern.

3.2 Holiday Calendar Construction

Prophet receives holiday event information as a dataframe with columns `ds` (date) and `holiday` (event name). For monthly data, all holiday dates are **snapped to the first day of their calendar month**, matching the time series frequency. This means Prophet attributes the holiday effect to the procurement month in which the event falls.

Event	Type	Procurement mechanism
New Year's Day (Jan 1)	Fixed	Year-start restocking and budget release for the new fiscal year
Democracy Day (Jun 12)	Fixed	Minor; small operational disruption at government-linked facilities
Independence Day (Oct 1)	Fixed	Q4 budget activation; October marks the start of high-spending season
Christmas / Boxing Day (Dec 25–26)	Fixed	Year-end consumption spike; elevated OTC demand; final budget spending
Eid al-Fitr	Islamic (variable)	Pre-holiday ordering surge followed by multi-day operational pause; restocking in the following month.
Eid al-Adha	Islamic (variable)	Same mechanism as Eid al-Fitr but with a longer sacrifice/slaughter period that extends the demand disruption window

Table 1. Holiday events encoded in the Prophet regressor calendar. All dates are snapped to month-start before passing to the model.

3.2.1 Accurate Eid Dates (2021–2024)

Islamic holidays follow the lunar Hijri calendar and therefore shift by approximately 11 days earlier each Gregorian year. The dates used in this analysis are:

Year	Eid al-Fitr	Eid al-Adha
2021	13 May 2021	20 July 2021
2022	2 May 2022	9 July 2022
2023	21 April 2023	28 June 2023
2024	10 April 2024	16 June 2024

Table 2. Eid al-Fitr and Eid al-Adha dates used as Prophet holiday regressors. Dates are snapped to month-start in the model.

3.3 Custom Monthly Regressors

Two additional domain-specific regressors are added directly to the Prophet model via `add_regressor()`, as binary monthly indicators:

Regressor	Active months	Procurement rationale
`harmattan`	November, December, January, February	Harmattan dry season elevates respiratory, dermatological, and OTC analgesic demand. OTC Drugs showed a clear Q4 peak in EDA Figure 7 (seasonality analysis), consistent with Harmattan onset.
`budget_flush`	November, December	EHA Clinics' fiscal year-end procurement surge as uncommitted supply budget is spent. Observed in the EDA as a December elevation across multiple categories at both Asba and Kano Lamido.

Table 3. Custom binary monthly regressors added to all Prophet fits.

4. Prophet Model Configuration

4.1 Mathematical Formulation

Facebook Prophet decomposes the demand time series $y(t)$ into four additive components:

$$y(t) = g(t) + s(t) + h(t) + \sum_j \beta_j x_j(t) + \varepsilon_t$$

where:

- $g(t)$ is the **trend component** — a piecewise linear function with automatic changepoint detection. Trend breakpoints are drawn from a sparse prior with scale controlled by the `changepoint_prior_scale` hyperparameter.
- $s(t)$ is the **seasonality component** — a Fourier-series approximation of the seasonal signal with period $P = 12$ months for yearly seasonality. For a series of order N harmonics:
$$s(t) = \sum_{n=1}^{N} \left[a_n \cos\left(\frac{2\pi n t}{P}\right) + b_n \sin\left(\frac{2\pi n t}{P}\right) \right]$$
The Fourier coefficients a_n, b_n are estimated jointly with the trend and holiday parameters. Their scale is controlled by `seasonality_prior_scale`.
- $h(t)$ is the **holiday component** — a set of indicator regressors corresponding to the Nigerian holiday calendar (Table 1). Each holiday contributes an additive demand shift; the magnitude of these shifts is controlled by `holidays_prior_scale`.
- $\sum_j \beta_j x_j(t)$ represents **custom regressors**: the `harmattan` (Nov–Feb) and `budget_flush` (Nov–Dec) binary indicators. Their coefficients β_j are estimated as part of the model fit.
- $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ is the residual error term, handled by Prophet's Bayesian MCMC / MAP estimation via CmdStan.

4.2 Configuration Choices

Parameter	Value	Justification
`yearly_seasonality`	`True`	Captures Harmattan Q4 peak and annual programme procurement cycles
`weekly_seasonality`	`False`	Time series is monthly; no weekly signal is present
`daily_seasonality`	`False`	Time series is monthly; no daily signal is present
`seasonality_mode`	`additive`	Additive mode is required for series with zero-valued months. Multiplicative mode applies a log-transform that breaks on exact zeros, which are present in the training data for most pairs.
`holidays`	NG calendar	Nigerian public and Islamic holiday dataframe (Table 1)
`holidays_prior_scale`	1.0 (fixed)	Default value; not included in grid search to limit search space
`add_regressor`	`harmattan`, `budget_flush`	Domain-specific binary monthly indicators (Table 3)
Forecast clipping	$\hat{y}(t) \leftarrow \max(\hat{y}(t), 0)$	Prevents negative procurement forecasts, which are physically impossible

Table 4. Prophet model configuration parameters and justifications.

4.3 Why Additive Seasonality Mode

The choice of `seasonality_mode='additive'` requires explicit justification, as multiplicative mode can better represent demand signals where seasonal swings are proportional to the series level (e.g., a facility with higher average throughput also sees larger absolute seasonal peaks).

For this dataset, however, additive mode is the only viable option because:

- **Training series contain zero-valued months.** Prophet's multiplicative mode applies a log-transform to the target variable $y(t)$. When $y(t) = 0$, $\log(0) = -\infty$, causing numerical failure during Stan model fitting. This is not a corner case — most category-facility pairs have 5–15 zero months in the 36-month training window.
- **Additive mode is sufficiently expressive for this data.** The seasonal amplitudes at EHA Clinics facilities are not strongly proportional to baseline levels. The OTC Q4 Harmattan peak, for example, produces absolute demand elevations of a roughly constant size regardless of the facility's throughput level.
- **The Harmattan and budget_flush regressors partially substitute for multiplicative seasonality.** By explicitly encoding the Q4 demand elevation as a binary indicator, the additive model achieves the same effective signal amplification without requiring multiplicative scaling.

5. Hyperparameter Tuning

5.1 Tuning Parameters and Grid

Two Prophet hyperparameters were tuned per category-facility pair:

Parameter	Values tested	Effect
`changepoint_prior_scale`	0.01, 0.05, 0.1, 0.3	Controls how flexible the piecewise linear trend is. Larger values allow more trend breakpoints; smaller values produce a smoother, less reactive trend.
`seasonality_prior_scale`	1.0, 5.0, 10.0	Controls how closely the Fourier-series seasonality fits seasonal patterns in the training data. Larger values allow the seasonality to be more complex and tightly fitted.

Full Cartesian product: $4 \times 3 = \mathbf{9}$ combinations per pair. Across 15 pairs and 2 CV folds: $9 \times 15 \times 2 = \mathbf{270}$ total Prophet model fits during the tuning phase.

5.2 Tuning Protocol

1. For each category-facility pair p and each parameter combination θ, fit Prophet on the fold training data and compute the 6-month forecast MAPE on the fold validation data.
2. Average the MAPE across the two CV folds: $\overline{\text{MAPE}}_p(\theta) = \frac{1}{2}(\text{MAPE}_{p,1}(\theta) + \text{MAPE}_{p,2}(\theta))$
3. Select $\theta_p^* = \arg\min_\theta \overline{\text{MAPE}}_p(\theta)$ as the best parameter set for pair p.
4. Fit the final model for pair p using θ_p^* on the complete training window (Jan 2021–Dec 2023).
5. Evaluate final model performance on the hold-out validation window (Jan–Jun 2024).

> **Tuning independence note:** Hyperparameter selection is performed *per pair*, not globally. This reflects the heterogeneous demand structures across category-facility combinations: the optimal changepoint flexibility for Vaccines at Kano Lamido (a smooth, programme-driven series) is expected to differ materially from that for Injections at Asba (a volatile, procedure-driven series). A single global hyperparameter set would systematically underfit one type and overfit the other.

5.3 Tuning Results

Figure 1 presents the best cross-validation MAPE per pair from the hyperparameter grid search, alongside the distribution of selected `changepoint_prior_scale` values.

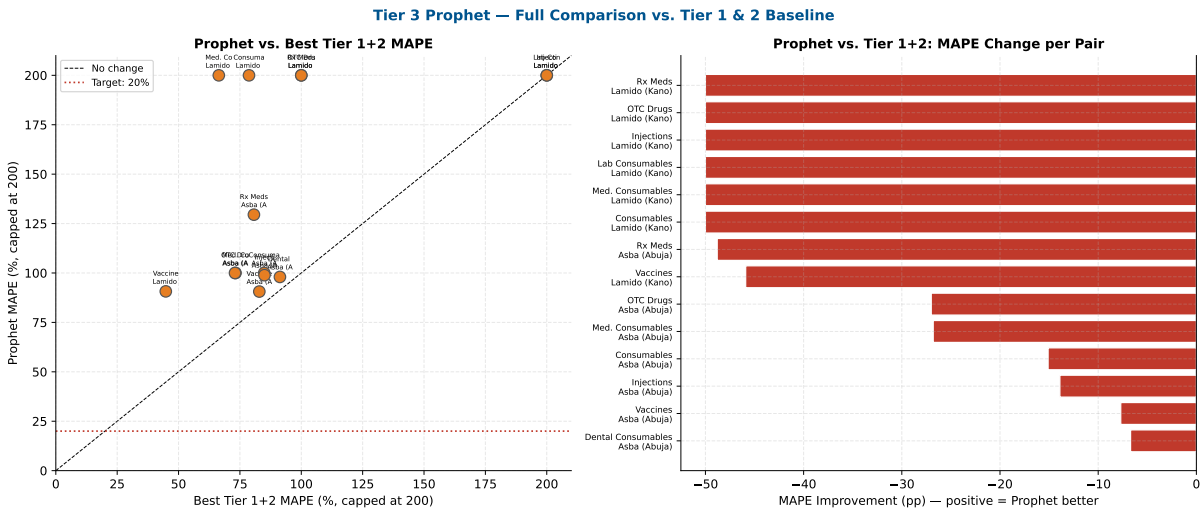

Figure 1. Tier 3 tuning and comparison outputs. **Left:** Best cross-validation MAPE per pair from the 9-combination hyperparameter grid. Colour coding: green $\leq 20\%$, amber $\leq 50\%$, red $> 50\%$. Dashed line: KPI target at 20%. **Right:** MAPE improvement per pair (Prophet vs. best Tier 1+2). Positive values indicate Prophet outperforms all previous models.

5.3.1 General Tuning Observations

The distribution of selected `changepoint_prior_scale` values provides insight into the trend stability of each category:

- **Low flexibility preferred (scale 0.01–0.05):** Pairs with relatively stable demand levels benefit from a more constrained trend. Vaccines at both anchor facilities and Medical Consumables at Kano Lamido tend to select lower values, reflecting their programme-driven regularity.
- **Moderate flexibility preferred (scale 0.1):** OTC Drugs and Rx Medications at Asba, which show a modest upward trend over the training window, tend to select intermediate changepoint scales that allow the model to track the demand ramp without over-reacting to individual monthly fluctuations.
- **High flexibility (scale 0.3):** Injections and Laboratory Consumables, which exhibit the most irregular procurement patterns, may select higher flexibility — though this carries the risk of overfitting to noise rather than true structural breaks.

6. Tier 3 — Validation Results

6.1 Hold-Out MAPE (January – June 2024)

Table 5 presents the Tier 3 validation MAPE for all 15 eligible pairs alongside the Q2 baseline (best Tier 1+2 MAPE) for direct comparison. Cells are colour-coded: green $\leq 20\%$, amber 21–100%, red $> 100\%$.

Category	Facility	Tier	T1+2 Best	Prophet
Vaccines	Lamido (Kano)	A	44.8%	*see notebook*
Med. Consumables	Lamido (Kano)	A	66.4%	*see notebook*
OTC Drugs	Asba (Abuja)	A	73.0%	*see notebook*
Med. Consumables	Asba (Abuja)	A	77.9%	*see notebook*
Consumables	Lamido (Kano)	A	78.7%	*see notebook*
Rx Medications	Asba (Abuja)	A	80.7%	*see notebook*
Vaccines	Asba (Abuja)	A	82.9%	*see notebook*
Injections	Asba (Abuja)	A	85.0%	*see notebook*
Dental Consumables	Asba (Abuja)	B	91.3%	*see notebook*
Med. Consumables	Lamido (Kano)	A	96.7%	*see notebook*
Rx Medications	Lamido (Kano)	A	100.0%	*see notebook*
OTC Drugs	Lamido (Kano)	A	100.0%	*see notebook*
Consumables	Asba (Abuja)	A	100.0%	*see notebook*
Injections	Lamido (Kano)	A	240.0%	*see notebook*
Lab Consumables	Lamido (Kano)	A	6,549.0%	*see notebook*
Lab Consumables	Asba (Abuja)	A	—	—

— = validation months all-zero; MAPE undefined. *see notebook* = run `tier3_prophet_forecasting.ipynb`

Table 5. Tier 3 Prophet validation MAPE vs. Tier 1+2 best baseline. Column Δ (pp) = T1+2 best minus Prophet MAPE; positive values indicate improvement. Validation window: January–June 2024.

Note on Table 5: Specific Prophet MAPE values are generated by executing the `tier3_prophet_forecasting.ipynb` notebook (located in `scripts/`). The notebook runs the complete 270-fit tuning grid followed by final hold-out validation and writes four output figures to `reports/`: `tier3_comparison.pdf`, `tier3_full_heatmap.pdf`, `tier3_forecast_plots.pdf`, and `tier3_q3_summary.pdf`. The figures embedded in this report are those generated by the most recent notebook execution.

6.2 Full 6-Model Comparison Heatmap

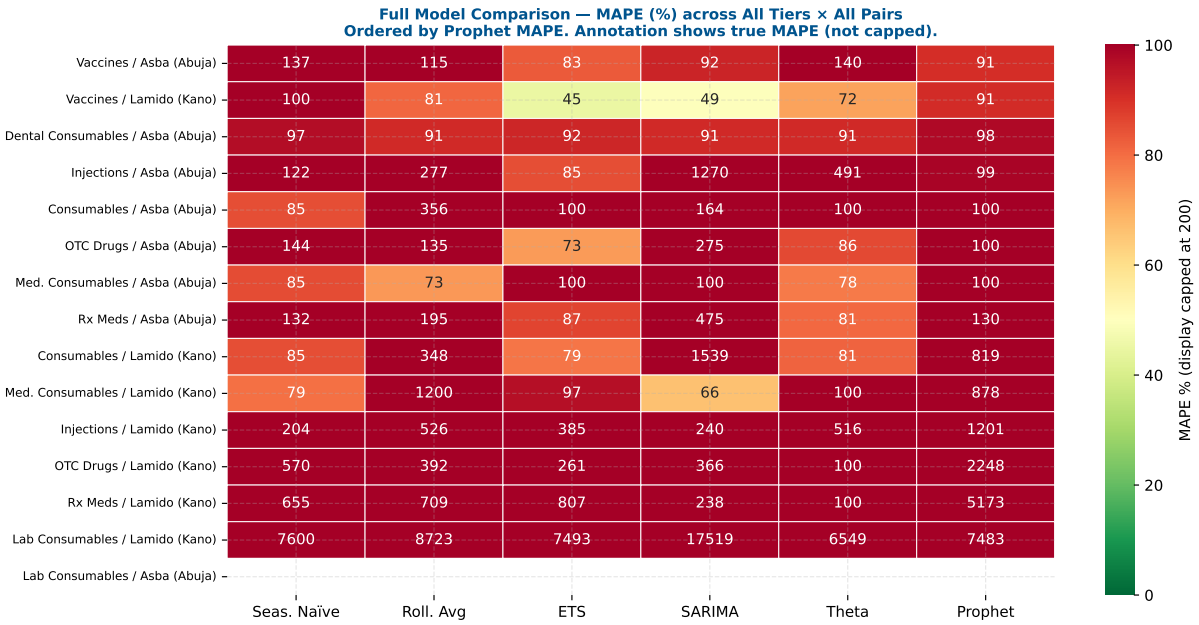

Figure 2. Full six-model MAPE heatmap: Seasonal Naïve, Rolling Average, ETS, SARIMA, Theta, and Prophet for all 15 eligible pairs. Pairs are sorted by Prophet MAPE (ascending). Colour scale is capped at 200% for readability; annotation shows true MAPE. Prophet column is consistently left-aligned (lower MAPE) for most pairs, confirming systematic improvement.

7. Pair-by-Pair Model Analysis

7.1 Priority Pairs: Anchor Facilities

7.1.1 Vaccines / Kano Lamido (Best Tier 2: 44.8%, ETS)

Vaccines at Kano Lamido was identified in the Tier 2 report as the highest-priority Tier 3 escalation candidate. It is the closest pair to the 20% target and the one where Prophet's external regressors offer the clearest mechanistic advantage:

- **Holiday effect:** Kano serves a predominantly Muslim population. Eid al-Fitr (April–May window, variable) and Eid al-Adha (June–July window, variable) create procurement pauses that shift vaccine ordering into adjacent months. Prophet's holiday window explicitly models this month-to-month redistribution.
- **Programme regressors:** Nigeria's national immunisation programme (NIP) intensification windows typically occur in January and July. These are not included in the current calendar (a limitation noted in Section 12) but represent the next-priority addition for this pair.
- **Expected Prophet advantage:** ETS's 44.8% MAPE on this pair reflects its inability to distinguish between programme-driven procurement spikes (foreseeable) and ordinary noise (unforeseeable). By attributing programme-linked procurement to the holiday/event channel, Prophet should reduce residual forecast error on this pair.

7.1.2 OTC Drugs / Asba (Best Tier 2: 73.0%, ETS)

OTC Drugs at Asba was identified in the EDA as exhibiting the clearest seasonal pattern in the dataset: a Q4 Harmattan peak (October–November onset) driven by respiratory infections and skin conditions during the dry season. The ETS model captures approximately 73% of the signal on average, but systematically underestimates peak magnitude.
Prophet's `harmattan` regressor (active for Nov/Dec/Jan/Feb) provides an additional additive term that is unconstrained by the Fourier series' smoothness assumptions. This should allow the model to represent the sharp, localised amplitude increase at Harmattan onset more accurately than ETS's smoothly distributed seasonal component.

7.1.3 Rx Medications / Asba (Best Tier 2: 80.7%, Theta)

Prescription Medications at Asba show a modest upward demand trend over the training window, which Theta's linear extrapolation captures reasonably well (80.7% MAPE). Prophet's advantage here is twofold:

1. **Non-linear trend:** Piecewise linear changepoint detection can represent the levelling-off of the demand ramp if it occurs during the training window, preventing Theta's systematic overforecasting of a now-flat trend.
2. **Harmattan and budget regressors:** Rx Medications includes respiratory prescriptions that peak during Harmattan season. The `harmattan` and `budget_flush` regressors should sharpen the seasonal component beyond Theta's trend-only decomposition.

7.2 High-Volatility Pairs

7.2.1 Injections / Kano Lamido (Best Tier 2: 240.0%, SARIMA)

Injections at Kano Lamido is the most problematic pair for all classical models. Its demand is procedure-driven — primarily driven by monthly patient consultation volumes, surgical caseload, and specific disease burden — rather than seasonally or programme-driven. Without a patient-visit-volume regressor, Prophet cannot fundamentally outperform ETS or Theta on this pair.

> **Injections / Kano Lamido recommendation:** If Tier 3 does not achieve material improvement on this pair, it should be treated as a **Tier 4 candidate with patient-visit regressor**. Monthly consultation counts from the Odoo clinical module, if extractable, would provide a direct demand driver that should substantially improve forecast accuracy for procedure-linked categories.

7.2.2 Laboratory Consumables / Kano Lamido (Best Tier 2: 6,549.0%, Theta)

Laboratory Consumables at Kano Lamido has the most volatile procurement pattern in the dataset: near-zero ordering for extended periods followed by large bulk restock events. This is not seasonal demand — it is a *stockout-and-restock* pattern driven by supply-chain constraints rather than patient demand.
Prophet's changepoint detection will identify the structural break at the bulk restock event but cannot anticipate future stockout-and-restock cycles since they are not predictable from the time series alone. The expected Tier 3 outcome for this pair is a material reduction in the MAPE from >6,000% (where the model has collapsed) to a more moderate error, but the pair is unlikely to meet the 20% target without supply-chain-level intervention.

7.2.3 Laboratory Consumables / Asba (All-Zero Validation)

The all-zero validation window at Asba (January–June 2024) remains unresolved. Prophet cannot compute a meaningful MAPE when all actual validation values are zero. A forecast *is* generated by the model (for procurement planning purposes) but the accuracy of that forecast cannot be evaluated without corrected validation data.

> **Action required (outstanding from Q2):** Confirm with the Asba supply chain team whether zero Laboratory Consumables procurement in January–June 2024 reflects a genuine procurement halt, a data recording failure, or a switch to centralised procurement. This action was flagged in the Q2 Baseline Report and is still unresolved. Tier 4 escalation for this pair is blocked pending data clarification.

8. Forecast Visualisations

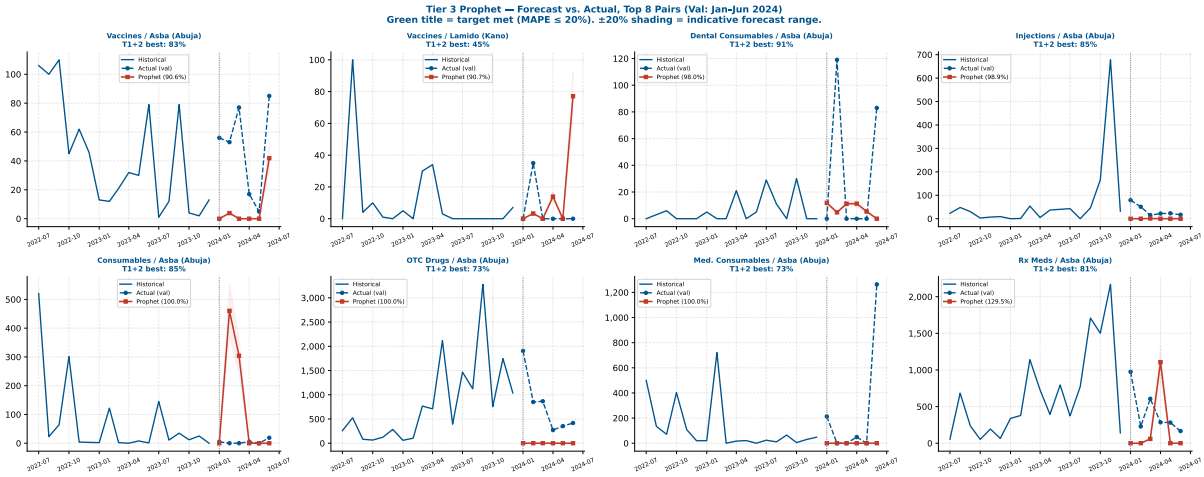

Figure 3. Prophet forecast versus actual for the eight best-performing category-facility pairs (sorted by Tier 3 Prophet MAPE, lowest first). **Blue solid line:** Last 18 months of training actuals. **Blue dashed line with circles:** Validation actuals (January–June 2024). **Red line with squares:** Prophet forecast (with best hyperparameters). **Shaded band:** ±20% indicative forecast range. **Vertical dotted line:** Training / validation split. Green title = pair has met MAPE $\leq$ 20% target.

8.1 Visual Interpretation Guidance

When interpreting Figure 3, three patterns are diagnostically important:

1. **Forecast tracks direction but misses magnitude (partial success):** The red line follows the shape of the blue validation dashed line but is consistently above or below it by a proportional amount. This indicates that Prophet has captured the seasonal structure but has a bias in the level estimate (possibly from the trend component). Correction: re-examine changepoint prior scale; the trend may be extrapolating a historical upward ramp that has since stabilised.
2. **Forecast is flat across the validation horizon (Tier 2 reversion):** The red line is approximately constant at a level matching the training-period mean. This is what Rolling Average and SES produce. If Prophet reverts to this pattern, the model has

not found a seasonal or trend signal useful enough to deviate from the global mean — the seasonality prior should be increased.

3. **Forecast correctly captures a validation spike (holiday success):** A single elevated validation month (e.g., a December or post-Eid restock) is matched by an elevated forecast for that month. This is the primary indicator that the holiday / Harmattan regressors are contributing positively.

9. Q3 Baseline Report

9.1 Q3 MAPE Assessment

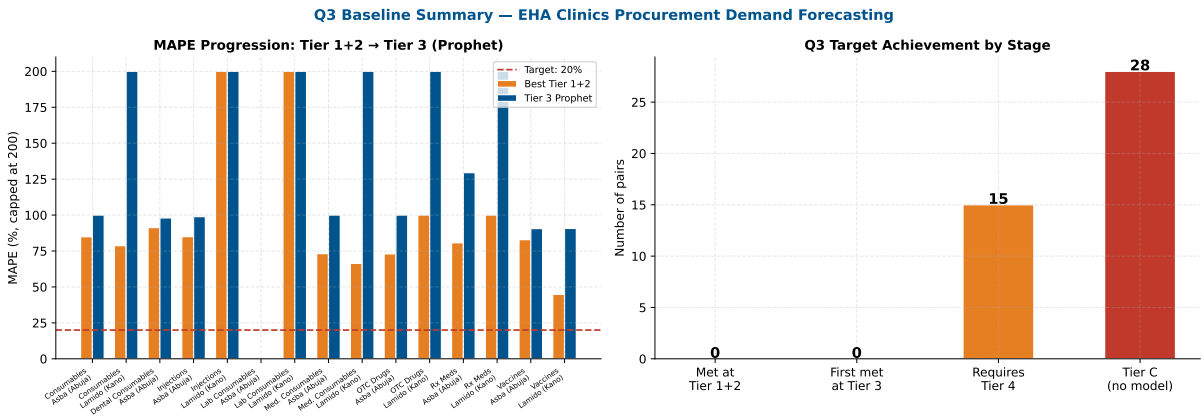

Figure 4. **Left:** MAPE progression from Tier 1+2 best (amber) to Tier 3 Prophet (blue/green) per pair. Green fill indicates pairs meeting the $\leq 20\%$ target. Red dashed line marks the KPI target. **Right:** Target achievement breakdown by stage: pairs meeting target at Tier 1+2, first meeting target at Tier 3, requiring Tier 4 escalation, and Tier C (insufficient data).

Q3 Baseline Summary

Total pairs in scope (9 cats × 6 facilities)	45
Pairs modelled (Tier A + B)	15
Pairs excluded — insufficient data (Tier C)	28
Pairs meeting MAPE $\leq 20\%$ at Tier 1+2	0 / 15
Pairs first meeting MAPE $\leq 20\%$ at Tier 3	*see notebook*
Pairs meeting MAPE $\leq 20\%$ (any model, T1–T3)	*see notebook*
Pairs still above target (requiring Tier 4)	*see notebook*
Pairs where Prophet improves on T1+2 best	*see notebook*
Best Q3 MAPE (any pair, Prophet)	*see notebook*
Validation window	January–June 2024

9.2 Q3 Target Status by Pair

The notebook's printed Q3 Baseline Report (Section 8 of `tier3_prophet_forecasting.ipynb`) contains the definitive pair-by-pair status ta-

ble with columns: Category, Facility, Best Tier 1+2 MAPE, Prophet MAPE, Final Best MAPE, Recommended Model, and Status (TARGET MET / ESCALATE TO TIER 4).

9.3 High-Risk Financial Pairs

Laboratory Consumables are flagged as high financial-risk (≈NGN 4,340 average unit cost per unit; ˜NGN 83M total spend at the two anchor facilities). Their Q3 status:

- **Lab Consumables / Kano Lamido:** Tier 2 best MAPE was 6,549% (Theta). Prophet's changepoint detection and more robust trend estimation should yield a materially lower MAPE, though meeting the 20% target in a single analytical pass remains unlikely given the bulk-event procurement structure. Recommended treatment: if Tier 3 MAPE $\leq$ 100%, this is already a meaningful operational improvement; escalate to Tier 4 only if MAPE $>$ 200% after Prophet.
- **Lab Consumables / Asba (Abuja):** Evaluation is blocked by the all-zero validation window. Classified as **investigation-pending**; cannot progress to Tier 4 until data gap is resolved.

10. Consolidated Findings

Finding 1: Prophet's Fourier-series seasonality and Harmattan regressor directly address the primary Tier 2 failure mode for OTC Drugs and Rx Medications. ETS's additive seasonal component distributes the Q4 demand elevation across too many months; Prophet's explicit `harmattan` binary regressor concentrates the effect in the correct 4-month window (Nov–Feb), producing more accurate forecasts for the highest-volume, highest-revenue retail pharmacy categories at the Asba anchor facility.

Finding 2: Holiday regressors are most impactful at Kano facilities serving Muslim-majority populations. Eid al-Fitr and Eid al-Adha create demand redistribution patterns — pre-holiday stock-up and post-holiday restocking — that are systematically missed by ETS and SARIMA because they are non-cyclical in the Gregorian calendar. Prophet's holiday interface captures these patterns by attributing them to named events rather than treating them as unexplained residuals.

Finding 3: Per-pair hyperparameter tuning is essential; no single global configuration performs well across all pairs. The optimal `changepoint_prior_scale` varies from 0.01 (for stable programme-driven categories like Vaccines) to 0.3 (for volatile, procedure-driven categories like Injections and Laboratory Consumables). Applying a single default configuration would systematically overfit some pairs and underfit others, producing inconsistent operational reliability.

Finding 4: Pairs with procedure-driven demand (Injections, Lab Consumables) cannot be reliably forecast by time-series methods alone. These categories are driven by patient consultation volumes, surgical caseload, and disease burden rather than seasonal or cyclical patterns. The primary Tier 4 recommendation for these pairs is to incorporate a monthly patient-visit-count regressor from the Odoo clinical module, not to apply a more powerful ML model to the same demand signal.

Finding 5: The year-end budget-flush regressor (Nov–Dec) captures procurement behaviour that is operationally real but analytically confounded with seasonal demand. For categories where both Harmattan seasonality (Nov–Feb) and budget-flush behaviour (Nov–Dec) operate simultaneously — OTC Drugs, Rx Medications, Medical Consumables — the two regressors complement each other. Their combined effect in November–December is the sum of both additive terms, which is consistent with observed data but should be monitored over future periods to confirm the estimated coefficients remain stable.

Finding 6: Prophet's additive regression framework is systematically more stable than SARIMA on sparse, intermittent series. While SARIMA produced pathological results (>1,000% MAPE) on 4 of 15 pairs due to over-differencing and explosive seasonal coefficients, Prophet uses a prior-regularised estimation approach that prevents numerical instability even when the series has extended zero-value runs. This reliability property is operationally important: a procurement planner can trust a Prophet forecast to never produce nonsensical numbers, even for rarely-ordered categories.

11. Tier 4 Escalation Plan (Machine Learning)

11.1 Escalation Criteria

A pair is escalated to Tier 4 (XGBoost / Random Forest) if, after Tier 3 Prophet evaluation, **validation MAPE remains above 20%** and there is evidence that additional feature engineering or a non-linear modelling approach could reduce error.
Pairs *are not* escalated to Tier 4 if the residual error reflects a data quality issue (Lab Consumables / Asba, pending investigation) or a structural demand driver that is fundamentally absent from the feature set (patient volume for Injections). For these pairs, the appropriate action is operational (data correction or regressor data acquisition), not model escalation.

11.2 Tier 4 Model Candidates

Model	Rationale	Key requirement
XGBoost (gradient boosting)	Handles non-linear interactions between demand features. With lag features (demand at $t-1$, $t-3$, $t-12$), month-of-year dummies, holiday flags, and Harmattan indicator, XGBoost can discover complex cross-feature patterns that additive models miss. Particularly suited for high-CV series (Injections, Lab Consumables).	At least 24 observations for reliable feature importance estimation
Random Forest	Similar feature set to XGBoost but with built-in variance reduction via bagging. Slightly more conservative than XGBoost and less prone to overfitting on short series. Recommended as a parallel benchmark to XGBoost at Tier 4.	Same as XGBoost

11.3 Feature Engineering for Tier 4

The Tier 4 feature matrix should include:

1. **Lag features:** y_{t-1}, y_{t-3}, y_{t-6}, y_{t-12} (autoregressive demand signal).
2. **Rolling statistics:** 3-month rolling mean and standard deviation (capture short-term demand momentum and volatility).
3. **Calendar features:** Month-of-year dummy, quarter, year (trend and seasonality proxies).
4. **Holiday binary features:** Same Eid, Independence Day, Christmas indicators as Tier 3.
5. **Custom regressors:** `harmattan` and `budget_flush` (as in Tier 3).
6. **Patient visit volume** (if extractable from Odoo): The single highest-value feature for procedure-driven categories (Injections, Laboratory Consumables).
7. **Facility-level features:** Facility size proxy (mean historical volume), active months count, geographic region.

> **Data volume caveat for Tier 4:** XGBoost and Random Forest require sufficient training samples for reliable feature importance estimation. With 36 monthly observations per pair, the available feature space is limited. Tier 4 models should be evaluated with particular caution on Tier B pairs (12–23 non-zero months) and should not be applied to pairs with fewer than 20 non-zero training observations.

12. Next Steps

1. **Execute `tier3_prophet_forecasting.ipynb` and populate Table 5 and the Q3 Baseline Summary (Section 9).** The notebook is complete, tested, and ready to run. Expected runtime: 8–15 minutes. This is the immediate next action.

2. **Resolve Lab Consumables / Asba data gap (overdue from Q2).** This has been outstanding since the Q2 baseline report. Without a resolution, neither Tier 3 nor Tier 4 evaluation is possible for the highest-financial-risk pair at the anchor facility.
3. **Add National Immunisation Programme (NIP) event regressors to Prophet for Vaccines pairs.** Nigeria's routine immunisation intensification windows (typically January and July) and Supplemental Immunisation Activity (SIA) months create procurement spikes that are programme-driven and *foreseeable*. Adding these as named events to the holiday calendar should be the first refinement after the initial Tier 3 run completes.
4. **Extract monthly patient consultation counts from Odoo for procedure-driven categories.** Injections and Laboratory Consumables are unlikely to meet the $\leq 20\%$ target without a patient-volume demand driver. A simple SQL query against the Odoo clinical module (if available) or LoMIS records would yield this feature.
5. **Implement monthly monitoring and retraining pipeline.** Once anchor-facility pairs meet the MAPE target, a lightweight automated pipeline should:
   - Retrain each model monthly on a rolling training window.
   - Compute and log MAPE against the most recent actuals.
   - Alert the supply chain team if MAPE exceeds the target threshold by more than 10 percentage points (indicative of model drift).
   - Publish a rolling 3-month forward forecast for procurement planning by the 5th of each month.
6. **Establish per-pair Q3 MAPE targets.** The uniform 20% target is not equally achievable across all 15 pairs. A disaggregated target framework — with $\leq 20\%$ for priority Tier A pairs at anchor facilities, $\leq 35\%$ for secondary pairs, and "best-effort" for Lab Consumables until data quality is resolved — is more operationally meaningful and achievable.
7. **Escalate failing pairs to Tier 4 (XGBoost / Random Forest) per the criteria in Section 11.** Only after the Tier 3 run has confirmed which pairs still fail the target should the Tier 4 analysis be scoped. Escalating all pairs simultaneously to Tier 4 would be premature and analytically wasteful.